NARASARAOPETA ENGINEERING COLLEGE (AUTONOMOUS)

Department of Computer Science and Engineering

Neuro-diagnosis: A YOLO-Inspired Pipeline for Automated Tumor Boundary Extraction and MRI-Based Classification and Segmentation

PRESENTED BY

Shaik Mohammad Khaleel (22471A0553)

Bontha John Babu (22471A0505)

Yannakula Shiva Sai (22471A0566)

Chowdari Rama Krishna (22471A0514)

Under the Guidance of,

K.V. Narasimha Reddy **M.Tech,**

Assistant Professor,

Department of Computer Science and Engineering,

Narasaraopeta Engineering College (Autonomous),

Narasaraopet- 522 601.

OUTLINE

1. Abstract
2. Introduction
3. Literature Survey
4. Research Gaps
5. Problem Statement
6. Objectives
7. Block Diagram / Flow Diagram
8. Methodology
9. Implementation
10. Results and Analysis
11. Conclusion & Future Scope
12. References
13. Question and Answers
14. Acknowledgements

ABSTRACT

The study focuses on **automated brain tumor detection** and **classification** from MRI images using **deep learning algorithms**, specifically **YOLO variants (YOLOv5, YOLOv7, and YOLOv8)**. After preprocessing the MRI datasets and training the models, the performance was compared using **precision, recall, and mean average precision (mAP)**. Among all, **YOLOv8 achieved the best results** with precision = 0.914, recall = 0.886, and mAP = 0.934 — showing its superiority in accurate and rapid tumor identification.

INTRODUCTION

In recent years, Artificial Intelligence (AI) and deep learning methodologies have shown significant promise in revolutionizing medical diagnosis by enabling automated, consistent, and highly accurate analysis of medical images. Among these, YOLO (You Only Look Once)–based object detection models have proven especially effective for real-time brain tumor identification in MRI images due to their speed and accuracy. YOLO performs detection and classification in a single forward pass, making it highly suitable for locating and identifying brain tumors efficiently and reliably.

LITERATURE SURVEY

No	Title	Author	Journal Name & Year	Methodology Adapted	Key Findings	Gaps
1	Automated Brain Tumor Detection and Localization Using YOLO	Hariharan P., Surya N.	*American Journal of Psychiatric Rehabilitation*, 2025	YOLO-based object detection on MRI scans	Achieved high tumor localization accuracy using YOLOv5	Limited dataset and no GUI integration
2	YOLO-TumorNet: An Innovative Model for Brain Tumor Detection	Huang J., Ding W., Yu G.	*Alexandria Engineering Journal*, 2025	Modified YOLOv7 with enhanced feature extraction	Improved detection of small and irregular tumors	Requires validation on multi-center datasets
3	PK-YOLO: Pretrained Knowledge Guided YOLO for Brain Tumor Detection	Kang M. et al.	*IEEE/CVF Winter Conf. on Applications of Computer Vision*, 2025	Multi-planar MRI analysis using pretrained YOLOv7	Enhanced accuracy in low-contrast MRI slices	Computationally expensive and time-consuming
4	Hybrid U-Net and YOLO Framework for Tumor Segmentation	Pehlivanoğlu M. K. et al.	*Eur. Phys. J. Spec. Top.*, 2025	Combined U-Net + YOLO for segmentation	Achieved better segmentation boundaries and localization	Complex architecture, high training time
5	SCC-YOLO:A Lightweight Model for Medical Image Detection	Bai R., Xu G., Shi Y.	*Int. Conf. on Health Big Data*, 2025	Spectral Channel Compression with YOLOv8	Reduced computation cost with high detection accuracy	Needs fine-tuning for 3D MRI volumes

RESEARCH GAPS

Lack of Integrated Models for Both Segmentation and Classification
Most existing approaches handle either segmentation or classification independently. There is a significant shortage of deep learning frameworks that **combine both tasks in a unified, real-time system**, which is essential for accurate and complete tumor analysis in clinical settings.

Insufficient Evaluation of YOLO Variants for Medical Imaging
Very few studies conduct a **comparative analysis of YOLOv5, YOLOv7, and YOLOv8** specifically for MRI-based brain tumor detection. This limits understanding of how architectural improvements across YOLO versions impact medical accuracy, speed, and reliability.

PROBLEM STATEMENT

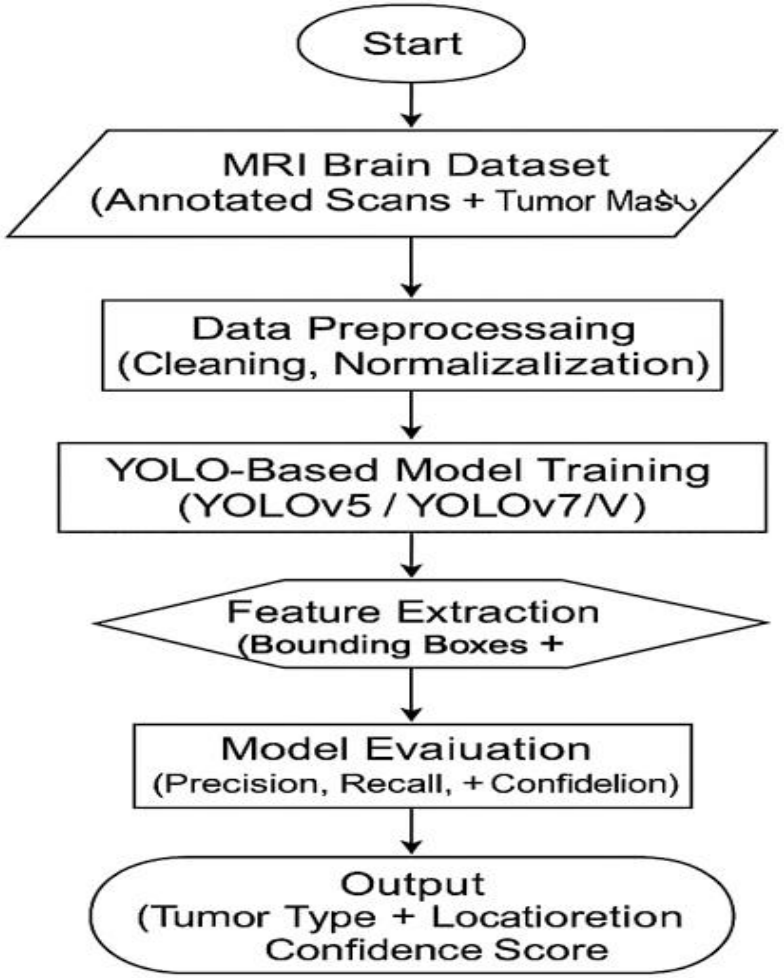
Accurate and early detection of brain tumors is challenging because manual MRI analysis is slow and error-prone, while traditional methods struggle with complex tumor features and lack real-time performance. To overcome these limitations, a reliable deep learning model is needed. This work aims to develop a YOLO-based system for automated brain tumor segmentation and classification to improve diagnostic accuracy and speed.

OBJECTIVES

1.To develop a YOLO-based deep learning model (YOLOv5, YOLOv7, YOLOv8) for accurate and real-time brain tumor detection, segmentation, and classification using MRI images.
2.To design an optimized preprocessing and model integration pipeline, including image enhancement, normalization, annotation alignment, and unified segmentation-classification workflow.
3.To evaluate and compare the model's performance using essential metrics like Precision, Recall, and mAP, and deploy it through a user-friendly GUI for real-time medical image analysis.

METHODOLOGY

This current research work involves in typical Deep Learning workflow simply as mentioned in the image side way that includes Data Collection, Data Preprocessing , model Training ,Evolution and prediction.
Each steps are furtherly elaborated in detailed in the upcoming slides



IMPLEMENTATION

Implementation:

Built **Google Colab** or model training and testing.

Dataset: **MRI Brain Dataset** containing annotated tumor masks (Glioma, Meningioma, Pituitary)

Software Specs:

OS: Colab

Programming Language: Python

Hardware Specs:

GPU: INTEL IRIS xe / (8 GB)

RAM: 8 GB | Storage: 100 GB (Google Drive)

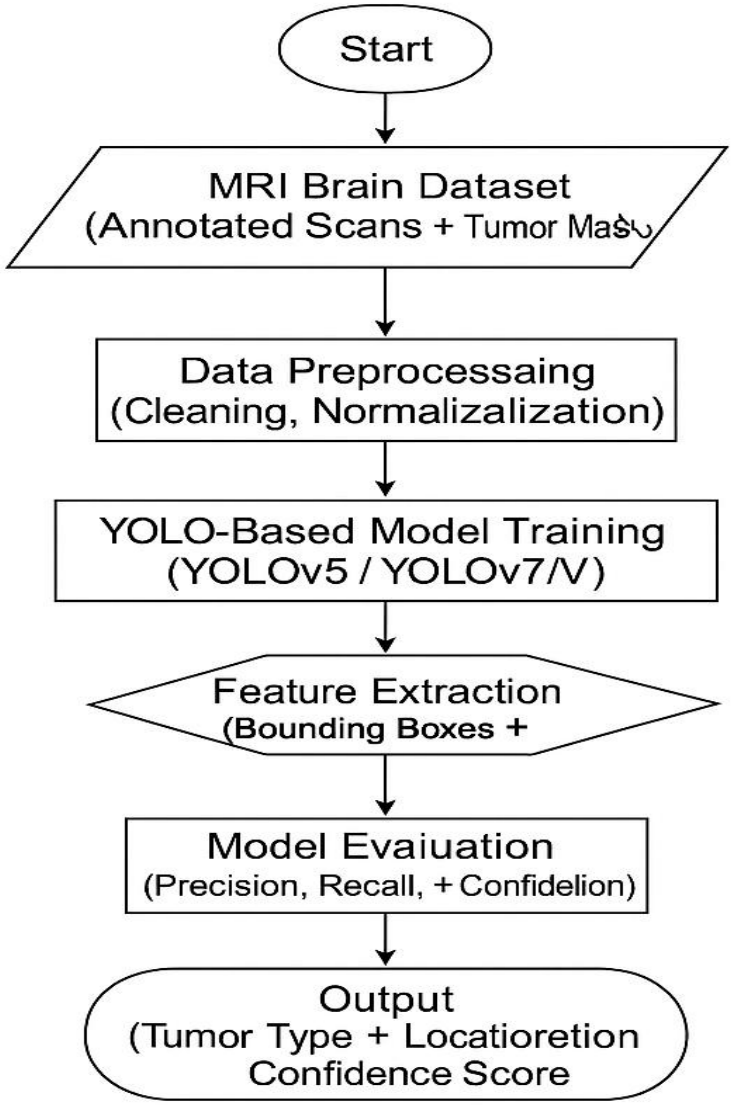
Challenges & Solutions:

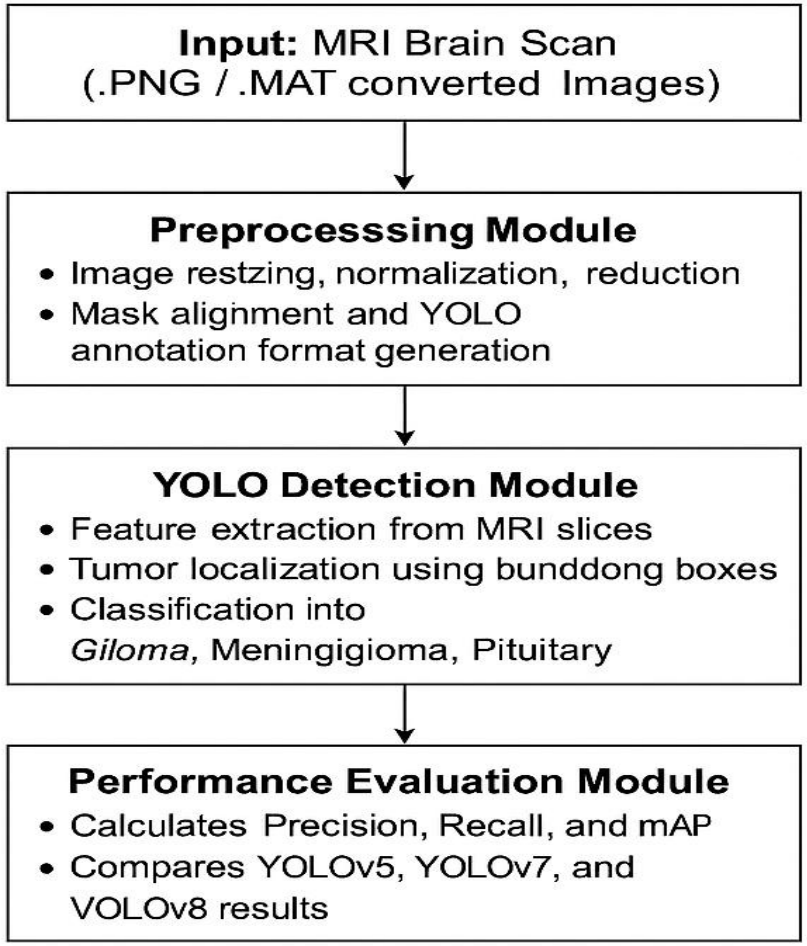
Data imbalance handled using augmentation.

Overfitting reduced with dropout and early stopping.

Real-time detection achieved via optimized YOLOv8 model.

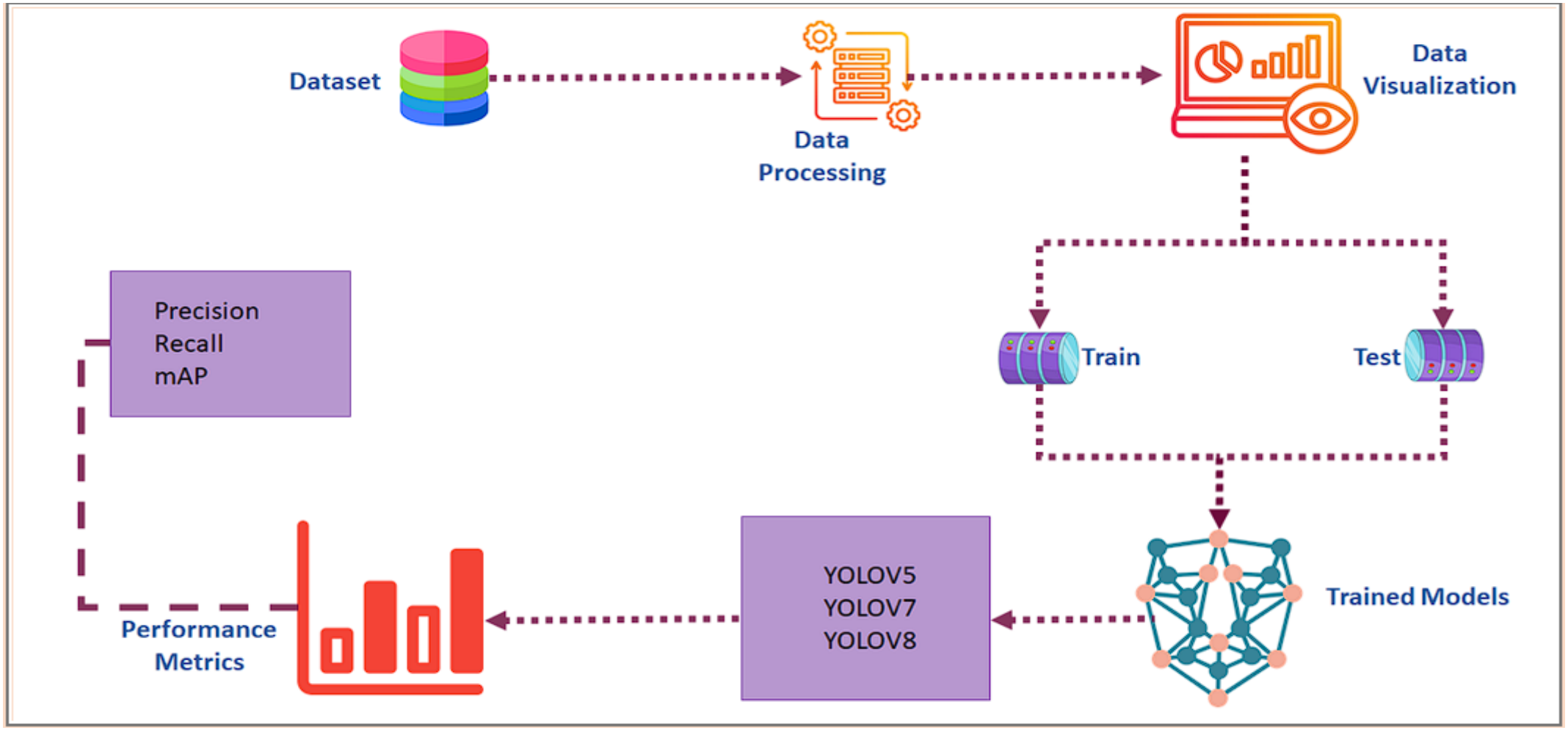
BLOCK DIAGRAM OR FLOW DIAGRAM





GUI Interface

SYSTEM ARCHITECTURE



PERFORMANCE EVALUTION TABLE

ML Model	Precision	Recall	mAP
Yolo v5	0.903	0.876	0.914
Yolo v8	0.914	0.886	0.934
Yolo v7	0.805	0.764	0.821

1.Model Comparison

Among all tested models, **YOLOv8** achieved the **highest performance** with **Precision = 0.914**, **Recall = 0.886**, and **mAP = 0.934**,outperforming YOLOv5 and YOLOv7 in both accuracy and detection reliability.

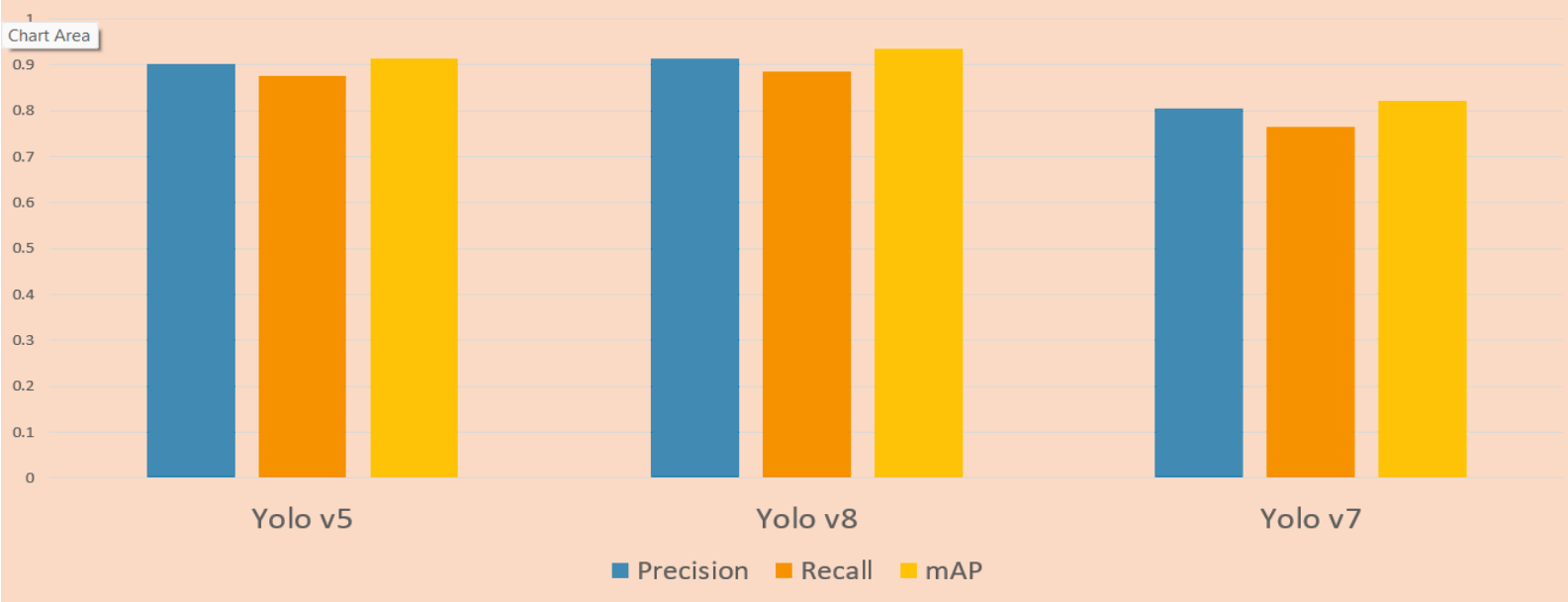
2.Detection Efficiency

YOLOv5 demonstrated balanced performance with fast training and stable accuracy,
while **YOLOv7** showed lower recall and struggled with small or irregular tumor shapes.

3.Overall Result

The experiment proved that **YOLOv8's improved architecture** (C2f + Transformer layers) significantly enhanced **feature extraction, localization, and classification**, making it the most **robust and clinically applicable model** for MRI-based tumor detection.

COMPARISON GRAPH



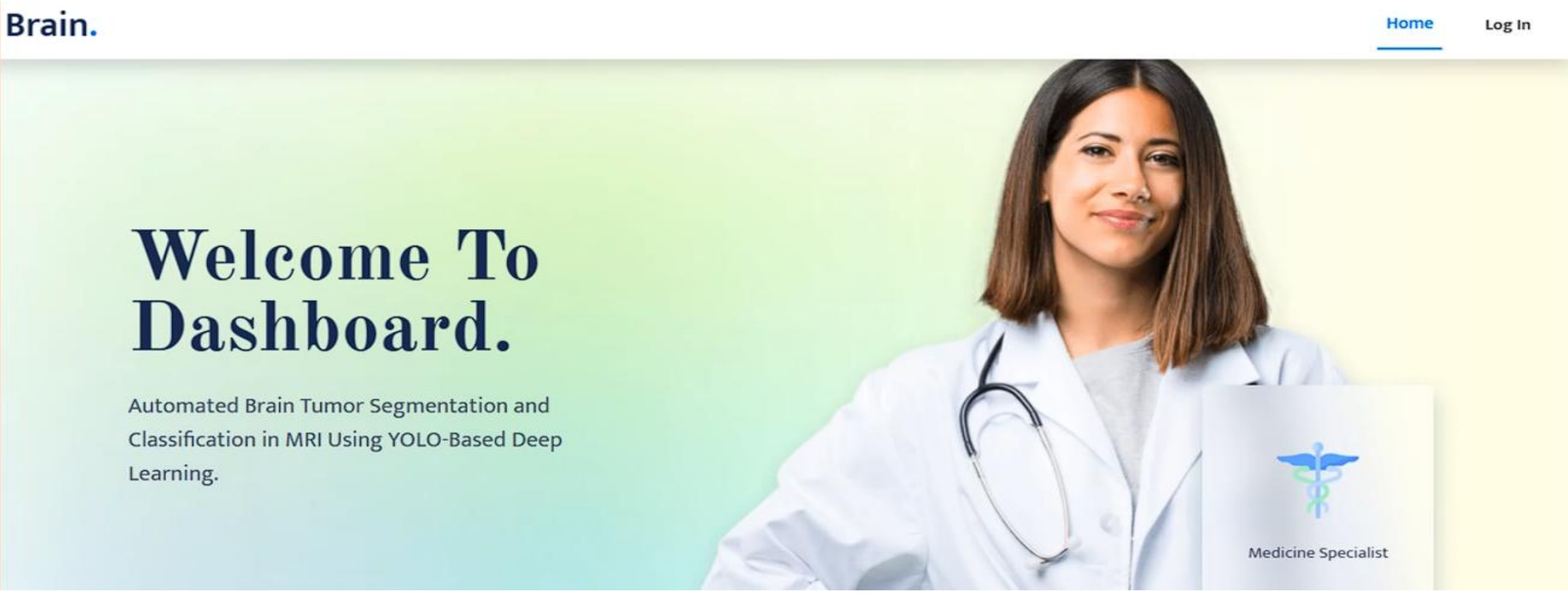
1.Model Accuracy Comparison

The graph clearly shows that **YOLOv8** achieved the **highest precision (0.914)** and **mAP (0.934)** among all models, indicating superior detection accuracy and localization capability.

2.Performance Consistency

YOLOv5 maintained stable precision (0.903) and recall (0.876), proving its reliability in balanced performance, while **YOLOv7** exhibited comparatively lower recall (0.764), showing reduced sensitivity in detecting small or complex tumor regions.

EXECUTION

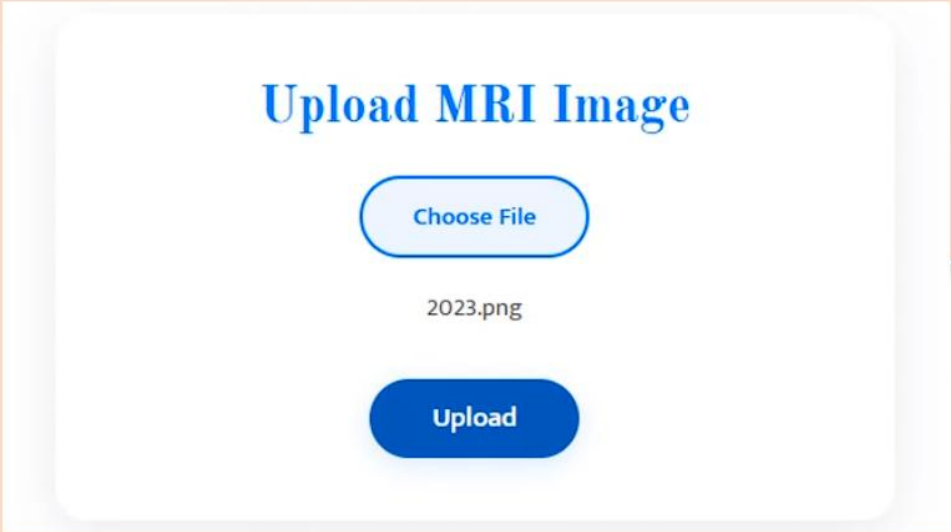

Brain.
Home
Log In
Welcome To Dashboard.
Automated Brain Tumor Segmentation and Classification in MRI Using YOLO-Based Deep Learning.
Medicine Specialist

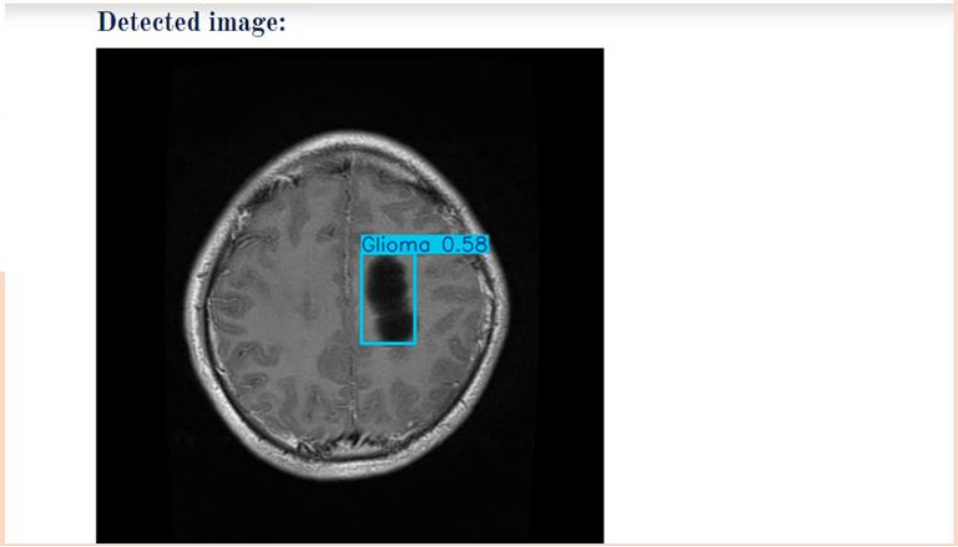
EXECUTION

Upload MRI Image
Choose File
174.png
Upload

Detected image:
Meningioma 0.92

EXECUTION


Upload MRI Image
Choose File
2023.png
Upload


Detected image:
Glioma 0.58

CONCLUSION

- The system successfully implements an automated brain tumor segmentation and classification framework using YOLO-based deep learning models. Through a systematic process involving data preprocessing, mask alignment, YOLO label generation, and visualization, the system ensures high-quality input for effective model training.
- Three different YOLO variants—YOLOv5, YOLOv7, and YOLOv8—were trained and evaluated to determine their effectiveness in detecting brain tumors from MRI images. Among these, YOLOv8 demonstrated the highest performance, achieving a precision of 91.4%, recall of 88.6%, and a mean Average Precision (mAP) of 93.4%.
- These metrics indicate a superior ability to accurately locate and classify tumor regions compared to the other models. The system is further enhanced by a user-interactive interface that allows users to input MRI images and receive real-time tumor detection results.

FUTURE SCOPE

- Future enhancements can focus on expanding the system to classify different types and grades of brain tumors for more detailed clinical insights. Incorporating 3D MRI data and volumetric analysis could improve spatial understanding and detection accuracy.

- Integration with larger and more diverse datasets will enhance model generalization. Additionally, embedding explainable AI (XAI) techniques can improve model transparency, aiding medical professionals in understanding predictions and building trust in AI-assisted diagnostic tools. Real-time deployment in clinical settings is also a potential direction.

REFERENCES

[1] P. Hariharan and N. Surya, “Automated brain tumor detection and localization using YOLO-based deep learning algorithm,” Am. J. Psychiatr. Rehabil., vol. 28, no. 5, pp. 198–203, 2025

[2] M. K. Pehlivanoglu, ˘ ˙I. ˙Ince, B. A. Kından, A. G. Eker, and ˙I. Dogan, ˘ “Towards advanced brain tumor segmentation: A novel hybrid architecture integrating UNet, FCN, and YOLO models on the newly introduced BTS-DS 2024 dataset,” Eur. Phys. J. Spec. Top., pp. 1–15, 2025

[3] M. Xiong, A. Wu, Y. Yang, and Q. Fu, “Efficient brain tumor segmentation for MRI images using YOLO-BT,” Sensors, vol. 25, no. 12, p. 3645, 2025

[4] D. Alamiri et al., “Applying deep learning to MRI image analysis for brain tumor classification,” in Proc. 6th Int. Conf. Bio-eng. Smart Technol. (BioSMART), 2025, pp. 1–5.

[5] M. K. Thakur and S. Chauhan, “Brain tumor detection and classification using YOLO algorithms,” in Proc. Int. Conf. Inventive Comput. Technol. (ICICT), 2025, pp. 116–123.

[6] M. H. O. Alnageeb and M. H. Supriya, “Real-time brain tumour diagnoses using a novel lightweight deep learning model,” Comput. Biol. Med., vol. 192, p. 110242, 2025.

[7] K. V. N. Reddy, V. R. Dodda, R. R. Mule, P. R. Babu, and M. Aparna, “Brain tumor detection using deep learning techniques,”AIP Conf. Proc., vol. 3162, p. 020072, 2025

[8] J. Huang, W. Ding, T. Zhong, and G. Yu, “YOLO-TumorNet: An innovative model for enhancing brain tumor detection performance,” Alexandria Eng. J., vol. 119, pp. 211–221, 2025

REFERENCES

[9] M. Kang, F. F. Ting, R. C. W. Phan, and C. M. Ting, "PK-YOLO: Pretrained knowledge guided YOLO for brain tumor detection in multiplanar MRI slices," in Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV), 2025, pp. 3732–3741.
[10] S. L. Jagannadham, K. L. Nadh and M. Sireesha, "Brain Tumour Detection Using CNN," 2021 Fifth International Conference on I-SMAC (IoT in Social, Mobile, Analytics and Cloud) (I-SMAC), Palladam, India, 2021, pp. 734-739, doi: 10.1109/I-SMAC52330.2021.9640875
[11] S. Khalighiyan, "Brain tumour identification through advanced machine learning techniques," M.S. thesis, Univ. Windsor, Canada, 2025.
[12] N. B. Bahadure, S. Routray, J. C. Patni, N. Raju, and P. D. Patil, "Berkeley wavelet transform and improved YOLOv7-based classification technique for brain tumor severity prediction," Int. J. Electr. Comput. Eng., vol. 15, no. 1, 2025.
[13] R. Bai, G. Xu, and Y. Shi, "SCC-YOLO: An improved object detector for assisting in brain tumor diagnosis," in Proc. Int. Conf. Health Big Data, 2025, pp. 114–120
[14] S. Uzun and M. Okuyar, "A new deep learning-based GUI design and implementation for automatic detection of brain strokes with CT images," Eur. Phys. J. Spec. Top., vol. 234, no. 1, pp. 141–164, 2025.
[15] S. Muksimova, S. Umirzakova, S. Mardieva, N. Iskhakova, M. Sultanov, and Y. I. Cho, "A lightweight attention-driven YOLOv5m model for improved brain tumor detection," in *Proc. IEEE Conf. Computers in Biology and Medicine*, vol. 188, p. 109893, 2025.